

Avinash K. Goyal

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EDUCATION

Northeastern University

2024 - 2028

- Candidate for **B.S. Mechanical Engineering**
 - Relevant Classwork: **Mechanics of Materials**, **Electrical Engineering**, **Dynamics**, & **Thermodynamics**
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SKILLS & CERTIFICATIONS

- Software: **SolidWorks (3D Modeling, FEA & Drawings/BOMs)**, AutoCAD, & Microsoft Suite
 - Electronics & Programming Languages: **ESP32 & Arduino Microcontrollers**, Soldering, Java, & C++
 - Machinery & Manufacturing: **3D Printer**, **CNC Machine**, Laser Cutter, Hand Tools, & Band Saw
 - Certifications: **SolidWorks Certified Associate**
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PROJECTS

Dual-Mode (Walking/Jogging) Transtibial Prosthetic | Personal Project

May 2026 - Present

- Ongoing project to design a **~15" prosthetic system integrating a walking foot and running blade**, switchable via a rotation and locking mechanism
- **Reviewed peer-reviewed biomechanics literature** to inform blade geometry optimized for jogging
- Performed **SolidWorks FEA** on the blade using Hexcel AS4C Carbon Fiber, **identifying stress concentrations** at the blade's curved segment and confirming a **factor of safety of 3.7**

Digital Puzzle Box | Personal Project

Sept - Oct 2025

- Designed and built an interactive, multi-stage puzzle box using an **ESP32**, **LCD**, **distance sensors**, **buttons**, **joystick input**, and **servo activation**
- Programmed 3 unique sequential logic puzzles in **C++ (Arduino IDE)**, including distance-based timing, binary decoding, and joystick input patterns to add variety
- Fabricated a functional prototype in under a two-week build timeline using **rapid, low-cost materials**

Plant Drum Spinner Combat Robot | Northeastern Robotics Club

May - June 2025

- **Designed and 3D printed** a compact, 1 lb combat robot featuring a drum spinner weapon system and low-profile chassis
- Designed a full **SolidWorks assembly** for chassis, armor, forks, and weapon geometry
- **Integrated electronic systems** within tight spatial constraints, including custom soldered circuitry

TransOptions Hydrogen Fuel Cell Car | Engineering Competition

Nov - Dec 2023

- **Designed and built a model car powered by a hydrogen fuel cell to travel 30 feet as fast as possible, achieving award-winning performance**
 - Developed a structured testing process isolating variables one at a time (gear ratio, wheel diameter, vehicle weight), using timed runs to guide each, **driving iterations with data rather than intuition**
 - Achieved top performance in the competition (30 ft in 8.538s average, achieving **nearly double initial speeds**), earning the **Best Engineered Car Award** for systematic design and optimization
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WORK EXPERIENCE

E.P. Heller Company Machining Intern | Madison, NJ

May - June 2024

- Managed bit loading for CNC production runs, **supporting continuous operation across 1–3 machines**
- **Interpreted technical blueprints** to disassemble, diagnose, replace components, and reassemble air-powered tools, restoring them to working condition